

Multimodal Trajectory Representation Learning for Travel Time Estimation

Zhi Liu
Zhejiang University of
Technology
College of Computer
Science and Technology,
Zhejiang Key Laboratory of
Visual Information
Intelligent Processing
Hangzhou, China
lzhi@zjut.edu.cn

Xuyuan Hu
Zhejiang University of
Technology
College of Computer
Science and Technology
Hangzhou, China
huxvyan@gmail.com

Xiao Han*
Zhejiang University of
Technology
College of Computer
Science and Technology,
Zhejiang Key Laboratory of
Visual Information
Intelligent Processing
Hangzhou, China
hahahenha@gmail.com

Zhehao Dai
Zhejiang University of
Technology
College of Computer
Science and Technology
Hangzhou, China
zhehaodai@outlook.com

Zhaolin Deng
Zhejiang University of
Technology
College of Computer
Science and Technology
Hangzhou, China
211123120032@zjut.edu.cn

Guojiang Shen
Zhejiang University of
Technology
College of Computer
Science and Technology,
Zhejiang Key Laboratory of
Visual Information
Intelligent Processing
Hangzhou, China
gjshen1975@zjut.edu.cn

Xiangjie Kong
Zhejiang University of
Technology
College of Computer
Science and Technology,
Zhejiang Key Laboratory of
Visual Information
Intelligent Processing
Hangzhou, China
xjkong@ieee.org

Abstract

Accurate travel time estimation (TTE) plays a crucial role in intelligent transportation systems. However, it remains challenging due to heterogeneous data sources and complex traffic dynamics. Moreover, traditional approaches typically convert trajectory data into fixed-length representations. This overlooks the inherent variability of real-world motion patterns, often resulting in information loss and redundancy. To address these challenges, this paper introduces the Multimodal Dynamic Trajectory Integration (MDTI) framework—a novel multimodal trajectory representation learning approach that integrates GPS sequences, grid trajectories, and road network constraints to enhance the performance of TTE. MDTI employs modality-specific encoders and a multimodal fusion module to capture complementary spatial, temporal, and topological semantics, while a dynamic trajectory modeling mechanism adaptively regulates information density for trajectories of varying lengths. Two self-supervised pretraining objectives, named contrastive alignment and masked language modeling, further strengthen multimodal consistency and contextual understanding. Extensive experiments on three real-world datasets demonstrate that MDTI consistently outperforms state-of-the-art baselines, confirming its

robustness and strong generalization abilities. The code is publicly available at: <https://github.com/City-Computing/MDTI>.

CCS Concepts

• Information systems → Spatial-temporal systems; • Computing methodologies → Artificial intelligence.

Keywords

Trajectory representation learning, Multimodal data mining, Travel time estimation

ACM Reference Format:

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1 Introduction

With the rapid development of connected vehicle technologies, vast amounts of trajectory-based sensor data are continuously generated and shared on the Web. Within this context, travel time estimation (TTE) [12] has emerged as a core task in traffic flow analysis [16, 27, 43, 44]. By transforming heterogeneous trajectory data into a unified indicator, TTE provides essential input for downstream decision-making processes such as route planning, traffic guidance, and can also support signal control [15]. TTE focuses on predicting the time required to traverse a specific route from an origin to a destination, while accounting for dynamic factors such as traffic conditions [34], road characteristics, and environmental

*Corresponding author.



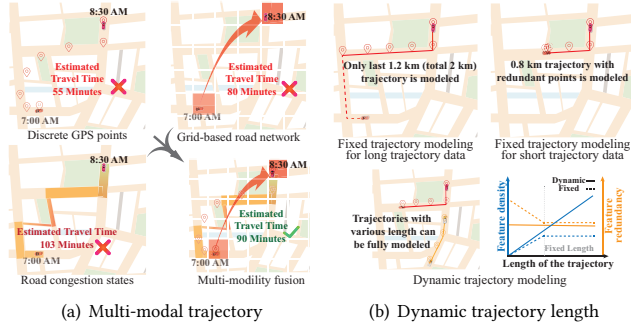


Figure 1: Toy examples of online trajectory data.

influences [25]. Accurate estimation enables more efficient route planning [5], congestion mitigation [13], and signal control [15] in urban transportation systems. Core aspects of TTE include its dependence on real-time data, integration of multimodal traffic information, and adaptability to rapidly changing traffic patterns. The main challenge, however, lies in effectively modeling the inherent complexity of multimodal spatiotemporal traffic systems. This underscores the need for advanced approaches to improve the precision and robustness of TTE.

Trajectory representation learning [23, 40] provides an effective framework for modeling multimodal spatiotemporal data, playing a crucial role in reconstructing complex trajectory patterns and supporting TTE tasks. Early trajectory representation learning (TRL) approaches mainly employed univariate sequence-to-sequence architectures [17, 36] to encode raw GPS trajectories [11, 14]. In these methods, trajectories are modeled as sequences of time-stamped data points, where each point contains a spatial coordinate pair (latitude and longitude), optionally accompanied by other attributes such as speed. Although such models effectively capture temporal dependencies and map sequential data into latent embeddings, they fall short in representing multimodal trajectory information, thereby limiting their applicability and robustness in real-world traffic scenarios.

As illustrated in Figure 1(a), discrete GPS trajectory data often suffers from drift (top left), grid-based road network modeling leads to limited accuracy (top right), and congestion-based analysis depends heavily on real-time updates (bottom left). Consequently, relying on a single-source trajectory data representation is insufficient for accurate and reliable travel time estimation. In contrast, integrating multiple information sources within a unified framework enables more precise and robust prediction of vehicle travel times, as demonstrated in the bottom right of Figure 1(a). Therefore, extracting semantic structures from different modalities and achieving effective alignment and fusion is a major challenge.

Another challenge in trajectory representation lies in converting dynamically varying trajectory samples into fixed-length feature vectors. For instance, GPS data collected from different vehicles often differ in spatiotemporal sampling frequencies [1], sequence lengths [39], and other factors. Many existing methods address trajectory variability by setting a maximum sequence length [18], padding shorter trajectories with zeros, and clipping longer ones for representation learning. However, these strategies often introduce redundant noise [8] or cause information loss. As shown in

Figure 1(b), when trajectories exceed the maximum threshold (top left), only the last segment is retained, preventing a full representation of the overall trajectory. Conversely, shorter trajectories (top right) are padded with redundant GPS points, which overemphasize local features such as the starting point and hinder effective representation learning. To overcome these issues, dynamic modeling techniques (bottom) can adaptively generate embeddings with differentiated information density and reduced redundancy as trajectory lengths [2, 6, 7] vary. Therefore, The second challenge is encoding trajectories of varying lengths and structures into uniform fixed-length feature representations.

To address these challenges, we propose the Multimodal Dynamic Trajectory Integration Model (MDTI), which jointly models GPS sequences, grid trajectories, and road network constraints to comprehensively represent trajectory features for travel time estimation. The proposed framework incorporates a multimodal fusion module that captures relationships among different modalities, enabling effective alignment and consistent semantic extraction. In addition, dedicated modality-specific encoders map heterogeneous inputs into a unified feature space, facilitating seamless multimodal fusion. Furthermore, a dynamic trajectory extraction module adaptively adjusts the information density of trajectory feature vectors according to trajectory length, preventing feature over-concentration in local embeddings and promoting balanced and discriminative representations. The main contributions of this work are summarized as follows:

- We propose the MDTI framework, which jointly leverages GPS sequences, grid trajectories, and road network constraints to systematically overcome the limitations of existing methods that rely on single or dual modalities, thereby enhancing semantic richness and completeness in trajectory representation.
- We designed a multimodal fusion module that not only aligns heterogeneous features and extracts consistent cross-modal semantics but also adaptively encodes trajectories of varying lengths, mitigating the inherent information loss and redundancy issues in fixed-length modeling.
- Extensive experiments on multiple real-world transportation datasets demonstrate that MDTI significantly outperforms state-of-the-art methods in TTE, validating the effectiveness of multimodal fusion and dynamic modeling in addressing the core challenges of trajectory representation.

2 Preliminary

Definition 1 (GPS Trajectory). A GPS trajectory is defined as an ordered sequence of raw GPS points collected at a fixed sampling interval by a GPS-enabled device:

$$\mathcal{T}^{gps} = \{p_i\}_{i=1}^T, \quad p_i = (x_i, y_i, t_i), \quad (1)$$

where x_i and y_i denote longitude and latitude, and t_i denotes the timestamp. Here, T is the total number of points in the trajectory.

Definition 2 (Grid Trajectory). A grid trajectory $\mathcal{T}^{grid}$ maps GPS points onto a discretized spatial grid. Let the region of interest be partitioned into $M \times N$ cells, each representing a spatial unit. The trajectory is then expressed as a sequence of grid cell identifiers:

$$\mathcal{T}^{grid} = \{g_t\}_{t=1}^T, \quad g_t \in \{1, \dots, M \times N\}, \quad (2)$$

where g_t denotes the identifier (ID) of the grid cell visited at time t .

Definition 3 (Road Network). A road network is represented as a directed graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V} = \{v_1, v_2, \dots, v_{|\mathcal{V}|}\}$ denotes the set of vertices, and each vertex $v_i \in \mathcal{V}$ corresponds to a distinct road segment. $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ denotes the set of directed edges, where each edge $e_{ij} = (v_i, v_j) \in \mathcal{E}$ represents the navigable connection from road segment v_i to v_j .

Definition 4 (Road Trajectory). A road trajectory $\mathcal{T}^{road}$ is a timestamped sequence of road segments obtained by map-matching GPS points $\mathcal{T}^{gps}$ to the road network $\mathcal{G}$. It can be expressed as

$$\mathcal{T}^{road} = \{v_t\}_{t=1}^T, \quad v_t \in \mathcal{V}, \quad (3)$$

where v_t denotes the road segment visited at time t .

3 Methodology

In this section, we present the proposed MDTI framework in detail. The overall architecture is shown in Figure 2. To handle multimodal trajectory data, three dedicated encoders—Grid Encoder, GPS Encoder, and Road Encoder—are designed, providing the model with preliminary multimodal processing capability. Specifically, graph attention and graph neural networks are employed to encode map-matched road trajectories, while large-model semantic analysis enhances the representation of raw GPS data. A multimodal fusion module is then introduced to align and integrate information across modalities, producing unified embeddings for TTE. In addition, we propose a dynamic trajectory alignment module that adaptively enriches the information density of embeddings, reducing redundancy and improving the accuracy of downstream TTE tasks. Finally, two complementary loss functions are applied to stabilize training and optimize performance. The following subsections provide a detailed description of each module in the MDTI framework.

3.1 Grid Encoder

Conventional grid-based modeling represents each cell and its internal trajectory points only by cell IDs, leading to sparse representations and limited adjacency awareness. To address this, we introduce a graph attention network (GAT)-based [37] embedding module that dynamically computes cell embeddings using input features such as speed and time. An 8-neighborhood connectivity scheme is adopted to better capture local spatial dependencies among grid cells.

Directly modeling a city-scale trajectory graph is computationally prohibitive. We therefore adopt a local subgraph strategy: for each trajectory point, we build a 3×3 grid-centered subgraph consisting of the current cell and its eight neighbors. This leverages the spatial locality of vehicle motion—single-step transitions rarely extend beyond adjacent cells—so the local neighborhood suffices to capture short-range dependencies while preserving trajectory continuity. Formally, for each node i , we retain edges only to itself and its eight immediate neighbors, yielding a first-order adjacency matrix A :

$$A_{ij} = \begin{cases} 1, & j = i \text{ or } j \in \mathcal{N}(i), \\ 0, & \text{otherwise.} \end{cases} \quad (4)$$

Here, $\mathcal{N}(i)$ denotes the 8-neighborhood of node i . This localized construction prevents the introduction of long-range, irrelevant

edges and ensures that trajectory modeling relies primarily on adjacent regions.

Given the graph structure $\mathcal{G}_{\text{grid}} = (X, A)$, each grid cell is treated as a node, with its channel vector serving as the initial node feature. Based on the adjacency matrix A , nodes are connected to their direct neighbors to form the local subgraph.

To capture spatial dependencies, the node features are encoded using a two-layer GAT. The GAT iteratively aggregates information from neighboring nodes according to the attention mechanism. For the l -th GAT layer, the embedding of node i is updated as:

$$h_i^{(l+1)} = \sigma \left(\sum_{j \in \mathcal{N}(i) \cup \{i\}} \alpha_{ij}^{(l)} W^{(l)} h_j^{(l)} \right), \quad (5)$$

where $h_i^{(l)}$ is the embedding of node i at layer l , $W^{(l)}$ is the learnable weight matrix, $\alpha_{ij}^{(l)}$ is the attention coefficient between nodes i and j , and $\sigma(\cdot)$ is a nonlinear activation function.

The attention coefficients are computed as:

$$\alpha_{ij}^{(l)} = \frac{\exp \left(\text{LeakyReLU} \left(a^\top \left[W^{(l)} h_i^{(l)} \parallel W^{(l)} h_j^{(l)} \right] \right) \right)}{\sum_{k \in \mathcal{N}(i) \cup \{i\}} \exp \left(\text{LeakyReLU} \left(a^\top \left[W^{(l)} h_i^{(l)} \parallel W^{(l)} h_k^{(l)} \right] \right) \right)}, \quad (6)$$

where a is the learnable attention vector and $\parallel$ denotes vector concatenation.

After two successive GAT layers, the node embeddings are aggregated into

$$H^{(2)} = \sigma \left(\text{GATConv}^{(2)} \left(\sigma \left(\text{GATConv}^{(1)}(X, A) \right), A \right) \right). \quad (7)$$

Before applying the projection, we aggregate the node embeddings of all grid cells into the matrix form H , which serves as the grid-level embedding map for subsequent temporal modeling. The embedding sequence is then projected as

$$G = \text{Proj}(H) \in \mathbb{R}^{T \times D}, \quad (8)$$

where $\text{Proj}(\cdot)$ denotes a linear projection, T is the number of temporal segments, and D is the Transformer input embedding dimension.

3.2 GPS Encoder

GPS data often exhibit spatiotemporal inconsistencies, including variations in timestamps, geographical locations, and sampling frequencies across trajectories. This heterogeneity poses major challenges for TTE, as conventional models struggle to capture inter-trajectory similarities and critical movement patterns from non-uniformly sampled data, resulting in reduced prediction accuracy. To address this issue, we explore the use of large language models (LLMs), leveraging their strong sequence modeling capabilities. However, applying LLMs directly to raw spatiotemporal trajectories faces a key obstacle: a modality gap between continuous coordinate values and the discrete token structure expected by LLMs, compounded by the absence of explicit semantic structure in raw GPS data. To bridge this gap, we propose a dedicated spatial discretization encoding method that maps continuous spatiotemporal coordinates into a sequence of discrete tokens.

Specifically, we decompose each trajectory into a sequence of structured textual segments with explicit semantics. Each trajectory is represented as a sequence of T time steps, where each time

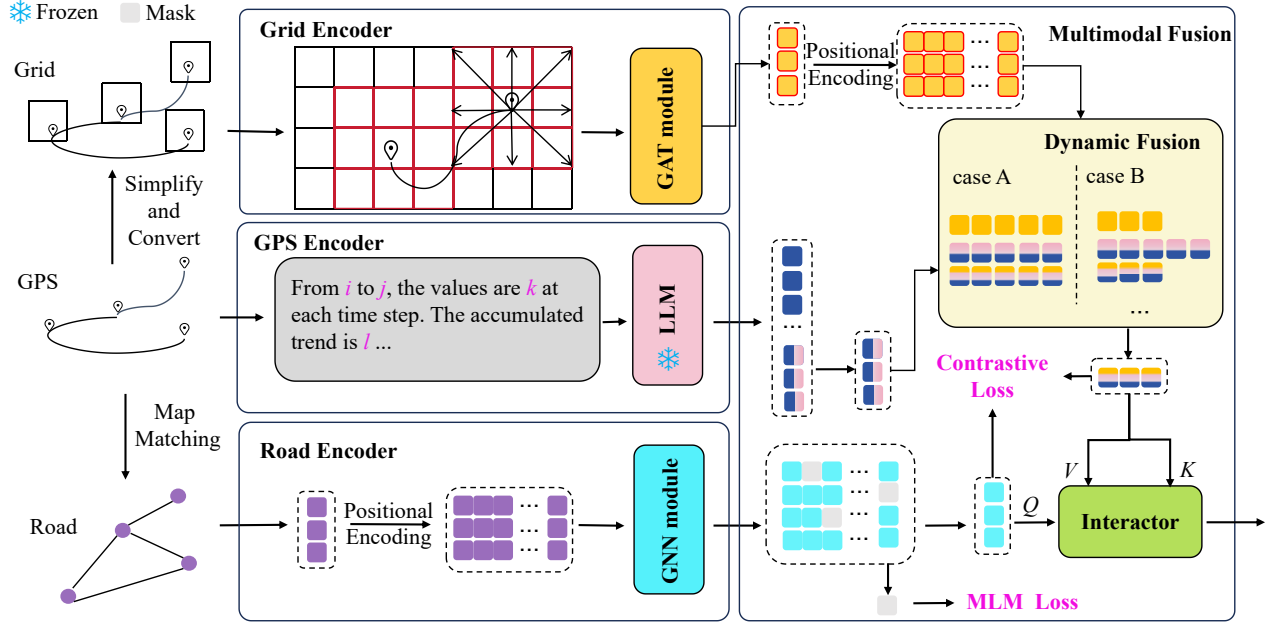


Figure 2: Overall architecture of the proposed MDTI framework.

step consists of three numerical channels, corresponding to the longitude, latitude, and temporal feature (e.g., timestamp or velocity). These channels jointly describe the spatiotemporal state of the moving object at each sampling point, serving as the fundamental input for local motion feature extraction. To extract local motion features, we divide the trajectory into non-overlapping chunks of length three, producing a set of sub-trajectories $\{\mathcal{S}_m\}_{m=1}^{T/3}$. The choice of length three is based on empirical observations [33]: it is long enough to capture local motion patterns while remaining computationally efficient for large-scale datasets. Each sub-trajectory $\mathcal{S}_m$ is flattened into a 9-dimensional vector $\tilde{p}_m$. To capture representative spatial-temporal motion patterns, we compare each $\tilde{p}_m$ with a pattern library $P = \{P_k\}_{k=1}^K \in \mathbb{R}^{K \times 9}$, where K denotes the number of predefined reference patterns. The similarity between $\tilde{p}_m$ and P_k is computed as:

$$S_{m,k} = \exp\left(-\frac{\|\tilde{p}_m - P_k\|_2}{2\sqrt{3}}\right). \quad (9)$$

The formula converts the similarities between each sub-trajectory and the prototypes into comparable weights by exponentiating the Euclidean distances, thereby identifying the most representative motion patterns. To further enhance semantic expressiveness, we also extract cumulative trend quantities from the original sequence:

$$\Delta = \sum_{t=2}^T (v_t - v_{t-1}), \quad (10)$$

where v_t represents the observed motion value at timestamp t .

After obtaining the structured motion information, we construct a natural-language prompt that integrates the temporal indices, cumulative trend, and pattern similarity results from Eq. 10. The prompt for the i -th trajectory segment is formulated as:

$$\text{Prompt}_i = f(\text{index}, \text{values}, \Delta, \text{patterns}, S_{i,k}), \quad (11)$$

where $f(\cdot)$ denotes the textual formatting function that converts numerical features into natural-language sentences. After feeding the prompts into a pre-trained language model, we take the embedding of the final token as the semantic representation of the current segment. Specifically, the embedding of the last token from the language model $\text{LM}(\cdot)$ output serves as the representation vector for the segment:

$$z_i = \text{LastToken}(\text{LM}(\text{Prompt}_i)) \in \mathbb{R}^{d_{\text{LM}}}. \quad (12)$$

This design combines prototype-based alignment, structured prompt generation, and representation learning with large-scale language models. It enables GPS trajectories to provide fine-grained motion semantics while leveraging the model's capacity to capture semantic dependencies and extract latent relationships across trajectories, thereby enhancing the overall framework's expressive power.

3.3 Dynamic Fusion

Traditional cross-attention fusion mechanisms enable interaction between modalities but suffer from inherent asymmetry: one modality dominates as the query source while the other serves as the key-value counterpart, leading to imbalanced fusion and difficulty

adapting to varying sequence lengths and structures. To address this, we propose a dynamic fusion mechanism that replaces the dominant-subordinate paradigm with symmetric and adaptive deep fusion, allowing bidirectional feature enhancement through fine-grained alignment and maximizing complementary information integration. In our framework, the trajectory representation generated by the GPS encoder acts as a prompt that guides and reinforces sequence learning in the grid encoder. Unlike conventional methods that simply concatenate or append GPS features to downstream predictors, our Dynamic Fusion module operates within the encoding stage, establishing early cross-modal correspondences between grid-based and GPS-based representations to ensure spatial and temporal consistency across modalities.

To guarantee that the two modalities can be effectively fused, we first apply a linear projection that maps GPS embeddings into the same latent space as the grid encoder output:

$$z'_t = \text{Proj}(z_t), \quad t = 1, \dots, T. \quad (13)$$

At the level of length, we introduce an alignment strategy to ensure consistent mapping between GPS sequences and grid tokens. When the GPS sequence is longer than the grid sequence, redundant temporal information is removed through truncation; when it is shorter, a zero-padding strategy is applied to preserve one-to-one correspondence. This mechanism achieves precise timestep synchronization across modalities and guarantees temporal consistency between GPS cues and grid tokens, providing a solid representation basis for subsequent fusion.

Finally, the aligned GPS sequence z' is injected into the grid sequence via a residual fusion mechanism:

$$\tilde{G}_t = G_t + z'_t, \quad t = 1, \dots, T, \quad (14)$$

where $\tilde{G}_t$ denotes the fused representation. This design preserves the spatial modeling capacity of the grid encoder while explicitly incorporating GPS prompts, thereby improving the consistency and alignment of the learned representations.

3.4 Road Encoder

The road modality provides topological constraints from the underlying road network, which are essential for TRL. Unlike GPS points that directly capture continuous spatio-temporal observations, road sequences are naturally embedded in a graph structure, where adjacency and accessibility are governed by the topology. To effectively capture such structured dependencies, we adopt a hybrid design: a GAT on the graph side to update road node embeddings, and a Transformer encoder on the sequence side to model temporal dependencies within trajectories.

On the graph side, each road node feature h_i is first projected into a unified embedding space through a linear transformation and then updated using stacked GATConv layers. The update process follows the same attention-based aggregation formulation as defined in Section 3.1 (Eq. 5–6).

Unlike the grid encoder, where the neighborhood $\mathcal{N}(i)$ is defined by fixed spatial proximity in a 3×3 grid, the road encoder constructs $\mathcal{N}(i)$ according to the topological connectivity of the road network. This allows message passing along physically connected road segments rather than adjacent spatial grids. Through this topology-aware aggregation, each node representation integrates

its own attributes and the structural dependencies of neighboring roads, producing road-aware graph embeddings.

On the sequence side, we construct the road sequence embedding table by augmenting three special tokens [PAD], [CLS], and [MASK]. Given a trajectory, we obtain the corresponding sequence embeddings and further add temporal embeddings (week index and minute index), type embeddings, and positional encodings $\text{PE}(\cdot)$ to preserve ordering. A padding mask ensures that attention is only applied to valid road segments.

The sequence encoder consists of stacked Transformer layers. In each layer, self-attention is first performed on the type embeddings, producing attention scores that are treated as adaptive priors B . These priors are then injected into the main self-attention computation:

$$\text{Attn}(Q, K, V) = \text{Softmax}\left(\frac{QK^\top}{\sqrt{d_k}} + B\right)V, \quad (15)$$

which explicitly incorporates structural priors into the trajectory modeling process. After multiple layers, the output sequence representations are projected into a shared space, and the first [CLS] token is taken as the trajectory-level representation. Finally, we denote the encoded road sequence representation as

$$\mathbf{R} = \{\mathbf{R}_t\}_{t=1}^T, \quad (16)$$

where $\mathbf{R}_t$ represents the contextualized embedding of the t -th road segment after GAT-based graph encoding and Transformer modeling. The overall trajectory-level representation is derived from the [CLS] token of $\mathbf{R}$.

3.5 Multimodal Interactor

A central challenge in spatiotemporal modeling lies in the effective fusion of heterogeneous modalities, such as GPS-based trajectory sequences and grid-based spatial features. Traditional fusion approaches (e.g., cross-attention [36]) often fail to establish fine-grained spatiotemporal correspondences, which is particularly detrimental for TTE. To overcome this, we design a modality interaction mechanism that leverages multi-head cross-attention to explicitly capture inter-modal dependencies, thereby mitigating sequence-length and sampling-frequency mismatches and achieving fine-grained semantic alignment across modalities.

Specifically, given a fused grid representation $\tilde{G} \in \mathbb{R}^{B \times L_g \times d}$ and a road sequence $\mathbf{R} \in \mathbb{R}^{B \times L_r \times d}$, we compute the cross-modal attention as:

$$\text{Attn}(\tilde{G}, \mathbf{R}) = \text{Softmax}\left(\frac{(\tilde{G}W_Q)(\mathbf{R}W_K)^\top}{\sqrt{d_k}} + M_s\right)(\mathbf{R}W_V), \quad (17)$$

where $W_Q, W_K, W_V \in \mathbb{R}^{d \times d_k}$ are learnable projection matrices, and M_s denotes the spatial alignment bias. In this design, the road sequence $\mathbf{R}$ acts as the *query*, actively retrieving relevant contextual information, while the fused grid representation $\tilde{G}$ serves as the *key* and *value*, providing rich spatial semantics. This formulation enables adaptive and context-aware feature interaction between modalities, guided by the trajectory structure.

The fused output is then refined through a residual connection followed by a feed-forward network:

$$\hat{X} = \text{FFN}\left(\tilde{G} + \text{Attn}(\tilde{G}, \mathbf{R})\right), \quad (18)$$

where $\hat{X}$ denotes the final fused representation. This feed-forward refinement enhances nonlinear feature transformation and improves the overall trajectory modeling accuracy.

3.6 Self-supervised Training

Contrastive Loss. To learn effective cross-modal representations between prompt-enhanced grid embedding and road network modalities, we employ a contrastive framework [32] that encourages alignment between semantically corresponding trajectory representations. Given a batch of trajectory pairs, where each sample contains both prompt-enhanced grid embedding visual representation and road network sequence representation, we aim to learn a shared embedding space where corresponding modalities are pulled together while non-corresponding pairs are pushed apart.

The contrastive learning objective employs InfoNCE loss with temperature scaling:

$$\mathcal{L}_{CL} = \frac{1}{2} [\mathcal{L}_{\tilde{G} \rightarrow R} + \mathcal{L}_{R \rightarrow \tilde{G}}], \quad (19)$$

where

$$\mathcal{L}_{\tilde{G} \rightarrow R} = -\frac{1}{N} \sum_{i=1}^N \log \frac{\exp(\tilde{G}_i \cdot \mathbf{R}_i / \tau)}{\sum_{j=1}^N \exp(\tilde{G}_i \cdot \mathbf{R}_j / \tau)}, \quad (20)$$

$$\mathcal{L}_{R \rightarrow \tilde{G}} = -\frac{1}{N} \sum_{i=1}^N \log \frac{\exp(\mathbf{R}_i \cdot \tilde{G}_i / \tau)}{\sum_{j=1}^N \exp(\mathbf{R}_i \cdot \tilde{G}_j / \tau)}. \quad (21)$$

Here, τ is a learnable temperature parameter that controls the concentration of the distribution, and N is the batch size. The bidirectional formulation ensures symmetric learning between both modalities.

Masked Language Modeling Loss (MLM Loss). To capture sequential dependencies and contextual relationships in road network trajectories, we adopt a BERT-style masked language modeling strategy [10]. In each trajectory sequence, 15% of the road segments are randomly masked, and the model is trained to recover the original road IDs based on surrounding context and cross-modal information from the *prompt-enhanced grid representations*. This design encourages the model to jointly leverage contextual and cross-modal cues while mitigating overfitting to the [MASK] token.

For a masked road trajectory $\tilde{s} = [s_1, s_2, \dots, [\text{MASK}], \dots, s_L]$, the model predicts the original road segment IDs at the masked positions. The training objective is defined as:

$$\mathcal{L}_{MLM} = -\frac{1}{|M|} \sum_{i \in M} \log P(s_i | \tilde{s}, \tilde{G}, \mathbf{R}), \quad (22)$$

where M denotes the set of masked positions, and $P(s_i | \tilde{s}, \tilde{G}, \mathbf{R})$ represents the probability of correctly predicting the road segment s_i given the masked sequence $\tilde{s}$ and its corresponding prompt-enhanced grid context.

This objective enables the model to learn rich contextual representations of road networks while leveraging spatial semantics from the prompt-enhanced grid modality, facilitating a deeper understanding of trajectory patterns and spatial dependencies.

Joint Training Objective. The final objective is the sum of the two self-supervised tasks:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{CL} + \mathcal{L}_{MLM}. \quad (23)$$

Both losses contribute equally, enabling the model to learn cross-modal alignment and intra-modal sequential patterns simultaneously.

4 Experiments

4.1 Experiment Settings

We evaluate the performance of MDTI on three real-world datasets. The experiments are designed to address the following research questions:

- **RQ1:** Does MDTI provide more accurate estimates of travel time compared to various baseline methods?
- **RQ2:** How do different components of the MDTI model contribute to the overall performance?
- **RQ3:** Can the dynamic trajectory modeling module of MDTI adapt to inputs of different trajectory lengths?
- **RQ4:** How is the transferability of MDTI?

Due to space limitations, we have placed RQ3 and RQ4 in the Appendix.

Datasets. We evaluate the proposed approach on three real-world multi-modal trajectory datasets collected from Porto, Chengdu, and Xi'an. Each dataset comprises GPS trajectories, route trajectories, and corresponding road networks. The GPS data are sourced from publicly available datasets released by Didi Chuxing¹, while the road networks are extracted using OSMNX [3], which provides rich attributes such as road type, length, lane count, and topological connectivity. For model training, we utilize only the topological structure of the road networks, differing from some baseline configurations that incorporate additional road attributes. The raw GPS trajectories are map-matched to the corresponding road networks through a standard map-matching algorithm, resulting in route trajectories and an assignment matrix. The assignment matrix explicitly encodes the mapping between trajectory sub-segments and their associated road segments. For data partitioning, all datasets follow a consistent split across cities: 60% for training, 20% for validation, and 20% for testing.

Implementation Details. Training is performed with a batch size of 32 for 30 epochs. We adopt Adam [20] as the optimizer with an initial learning rate of 2×10^{-4} , cosine annealing schedule, 10 warm-up epochs, a minimum learning rate of 1×10^{-6} , and weight decay of 1×10^{-4} . Both hidden and output embedding dimensions are fixed at 256 with a dropout rate of 0.1. For GPS trajectories, each sequence is segmented into patterns and compared with historical keys, and the most relevant ones are transformed into natural language prompts. These prompts are encoded by a pre-trained GPT-2 model, where the hidden state of the last token is taken as the trajectory embedding. Grid and road representations are generated by Transformer and GAT layers, while the cross-modal module aligns multi-source features. The objective combines multiple loss terms with weights of relation=0.5, step=1.0, rel=1.0, and cos=1.0.

Evaluation Metrics. We adopt two widely-used regression metrics: Mean Absolute Error (MAE) for measuring average prediction error magnitude, Mean Absolute Percentage Error (MAPE) for assessing relative error percentage. These metrics collectively provide a comprehensive assessment of prediction accuracy and model robustness.

¹<https://outreach.didichuxing.com/>

Baselines. We compare the proposed MDTI model with eight baseline methods, which can be categorized into four groups. All methods are trained using the same number of trajectories. The four groups include: one method based on GPS trajectories (e.g., Traj2vec [40]); one method based on grid trajectories (e.g., TrajCL [4]); four methods based on road trajectories (e.g., JCLRNT [31], START [18], JGRM [29], GREEN [45]); and two methods related to travel time estimation (e.g., DutyTTE [30] and MulT-TTE [24]). Further details about the baselines are provided in Appendix A.2.

4.2 Performance Comparison (RQ1)

Table 1 presents the overall performance of our model and baseline methods across the three city datasets. Several key observations emerge: **(1) Our method consistently achieves the best MAE on all datasets and the lowest MAPE on Chengdu and Xi'an.** This confirms that the proposed cross-modal interaction module and modality-specific encoders within a unified feature space effectively enhance cross-domain alignment and semantic representation. In addition, the dynamic trajectory feature extraction mechanism adaptively balances information density across trajectories of varying lengths, mitigating local embedding bias. **(2) Among single-modality trajectory modeling methods, Traj2vec—which directly encodes raw GPS point sequences—performs the worst, as RNN-based sequence modeling amplifies local noise.** In contrast, TrajCL, which relies solely on grid representations, suffers from spatial discretization errors and the loss of fine-grained positional information. Road-network-based methods generally outperform both, indicating that the topological connectivity of road segments and intersections provides strong structural constraints, while road semantics and traffic rules capture more stable spatiotemporal patterns. **(3) On the Porto dataset, our method performs slightly worse than MulT-TTE in terms of MAPE, likely due to MAPE's sensitivity to relative errors.** Nevertheless, our approach achieves the lowest MAE, demonstrating superior robustness to outliers and measurement noise. DutyTTE shows the weakest overall performance, as it lacks multimodal alignment and focuses primarily on global path consistency and interval coverage. **(4) The cross-modal trajectory representation learning methods outperform single-modality ones, confirming that multimodal information effectively compensates for modality-specific limitations.** Our dynamic fusion mechanism adaptively regulates information density according to trajectory length, preventing feature over-concentration in local segments. This design yields consistently stable performance across both short and long trajectories while reducing the impact of extreme values.

4.3 Component Contribution (RQ2)

Ablation Settings. To assess the contribution of each MDTI module, we conduct six ablations:

- **w/o GPS Enc.** Removes the GPS modality and the associated large language model, retaining only the road network and grid representations.
- **w/o Road Enc.** Removes the road network encoder, leaving only the GPS and grid encoders.

Table 1: Performance comparison across three datasets. The best results are in bold and the second-best are underlined. ↓ means lower is better. Statistical significance was tested using paired t-tests across datasets; the differences were significant with $p < 0.05$.

Method	Porto		Xi'an		Chengdu	
	MAE↓	MAPE↓	MAE↓	MAPE↓	MAE↓	MAPE↓
GPS-based trajectory representation learning						
Traj2vec	4.027	0.534	1.651	0.352	2.481	0.267
Grid-based trajectory representation learning						
TrajCL	2.085	0.236	1.632	0.321	1.609	0.536
Travel Time Estimation Models						
DutyTTE	3.780	0.303	4.131	0.302	4.525	0.316
MulT-TTE	1.524	0.147	<u>1.231</u>	<u>0.172</u>	1.461	0.234
Road-based trajectory representation learning						
START	1.541	0.173	1.764	0.282	1.359	0.218
JCLRNT	3.065	0.327	1.638	0.272	1.352	0.241
JGRM	2.993	0.375	1.457	0.333	1.374	0.360
GREEN	<u>1.510</u>	0.165	1.241	0.184	<u>1.322</u>	<u>0.185</u>
MDTI	1.419	<u>0.152</u>	1.169	0.170	1.143	0.183

- **w/o Grid Enc.** Removes the grid modality; the embeddings generated by the GPS encoder are fused directly with the road network features.
- **w/o Dynamic Fusion.** This variant removes the prompt-based Dynamic Fusion mechanism.
- **w/o CL Loss.** Removes the contrastive loss.
- **w/o MLM Loss.** Removes the MLM loss.

The outcomes of the ablation study conducted on the Porto dataset, as reported in Table 2, quantitatively validate the contribution of each component within the MDTI framework.

Importance of Multimodal Encoders. Removing any single modality encoder results in a consistent performance decline across all evaluation metrics (MAE and MAPE), highlighting the necessity of multimodal information fusion. Notably, the exclusion of the Grid Encoder yields the largest performance drop, indicating that grid-based spatial representations provide crucial contextual cues that complement the sequential dynamics captured by GPS trajectories and the topological structure encoded by the road network. The observed degradation when removing either the GPS Encoder or the Road Encoder further confirms that both trajectory dynamics and road topology contribute distinct and indispensable information for accurate TTE.

Effectiveness of Pre-training Objectives. Ablating the two self-supervised pre-training tasks also confirms their essential roles. Removing the MLM loss leads to a more pronounced degradation than removing the CL loss, suggesting that the MLM task—which reconstructs masked road segments based on contextual and grid information—plays a key role in modeling intra-modal dependencies and cross-modal interactions. The relatively smaller impact of removing the CL loss indicates that while inter-modal contrastive alignment enhances representation consistency, the model can still acquire meaningful embeddings through the MLM objective and supervised training. Nevertheless, the best results are achieved

when both objectives are jointly optimized, demonstrating their complementary effects in improving representation quality and downstream performance.

Indispensability of the Dynamic Fusion Alignment Mechanism. The experimental results clearly demonstrate that the proposed dynamic fusion alignment mechanism is crucial for maintaining the overall performance of the model. Its removal leads to a substantial drop across all evaluation metrics, producing outputs markedly inferior to those of the full model that benefits from dynamic feature alignment. Although some global metrics may appear comparable, qualitative inspection reveals evident spatial distortions and implausible artifacts in the predicted distributions. These findings confirm that the primary contribution of the dynamic fusion alignment mechanism lies in its ability to intelligently align and integrate multi-source features, ensuring spatial coherence and physical plausibility in model predictions, rather than merely optimizing global numerical accuracy.

Table 2: Ablation study of different modules.

	MAE↓	MAPE↓
w/o GPS enc.	1.441	0.153
w/o Road enc.	1.468	0.156
w/o Grid enc.	1.593	0.180
w/o CL loss	1.513	0.165
w/o MLM loss	1.540	0.167
w/o Dynamic FusionAlign	1.457	0.154
MDTI	1.419	0.152

5 Related Work

5.1 Multimodal Trajectory Representation Learning

TRL encodes sequences of states, actions, or spatiotemporal points into compact vectors that preserve temporal structures and semantic information. Early work introduced Traj2vec [40], which directly modeled raw GPS trajectories as point sequences to derive compact representations for downstream tasks. However, due to inherent issues in GPS data—such as positioning drift, U-turns, and redundant sampling—these approaches often produced unstable and poorly transferable representations [22].

To overcome these limitations, grid-based trajectory modeling methods have been proposed [4, 23]. By discretizing the spatial domain into grid cells and constructing graph-based structures, these approaches enhance the capacity for semantic and structural trajectory modeling. For instance, TrajCL [4] adopts contrastive learning to model spatial correlations and structural relationships among trajectories jointly.

In parallel, road-based trajectory modeling has emerged as another major direction. Using map-matching techniques [28], trajectories are aligned with road networks, which introduces topological constraints and enables richer representations of travel patterns. JCLRNT [31] generates contrastive samples between roads by leveraging contextual neighbors and constructing trajectory-level contrasts through detour path replacement. START [18] incorporates temporal regularities and travel semantics into a self-supervised

framework, improving both representation quality and generalization. JGRM [29] further strengthens spatiotemporal road features by mapping raw GPS points to corresponding road segments [19, 42, 46]. Based on this trend, GREEN [45] integrates grid-based and road-based representations to exploit their complementarity.

Building on these works, we unify GPS-based, grid-based, and road-based information in a multimodal framework to mitigate single-modality limitations for TTE.

5.2 Travel Time Estimation

Travel time estimation refers to predicting the time required to traverse a route or road segment from origin to destination. Existing studies can be broadly categorized into path-based methods and origin–destination (OD)-based approaches.

Path-based methods [21, 35] use complete trajectories or trajectory segments as input, enabling the modeling of vehicle dynamics within the road network. These approaches typically leverage road segment features, trajectory sequences, or path structures to capture spatiotemporal dependencies. For instance, DeepTTE [38] integrates graph convolutional networks with LSTM [41] to model the spatiotemporal dependencies of GPS sequences. HierETA [9] introduces a multi-view hierarchical choice model for fine-grained and interpretable TTE. MulT-TTE [24] jointly models trajectory, attribute, and semantic sequences, and further applies self-supervised learning to improve generalization, yielding significant gains in prediction accuracy.

OD-based methods [26, 30], by contrast, rely only on OD pairs and departure times, avoiding the need for full trajectory inputs. More recently, DutyTTE [30] combines reinforcement learning with a mixture-of-experts framework to optimize path prediction and capture segment-level uncertainty, enabling confidence interval estimation.

Our work is more closely aligned with path-based approaches, as it exploits the spatiotemporal information embedded in complete trajectories to achieve more accurate and robust TTE.

6 Conclusion

This paper proposes a multimodal dynamic trajectory representation framework, **MDTI**, which integrates GPS sequences, grid trajectories, and road network constraints to learn diverse trajectory representations. Corresponding encoders are designed for these three types of trajectories. In addition, to address the limitations of fixed-length feature vectors, a dynamic fusion module is developed. Finally, two loss functions are designed to enhance the framework's performance. Experiments on three datasets demonstrate that MDTI outperforms state-of-the-art baseline methods.

7 Acknowledgment

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A Appendix

A.1 Notation

We summarize and list all notations used throughout this paper in the following Table 3.

Table 3: Notation definitions used in this paper.

Notation	Description
$\mathcal{T}^{\text{gps}}$	an ordered sequence of raw GPS points
$\mathcal{T}^{\text{grid}}$	a grid trajectory
$\mathcal{G}$	a road network
$\mathcal{V}$	the set of vertices
$\mathcal{E}$	the set of directed edges
$\mathcal{T}^{\text{road}}$	a road trajectory
A	a first-order adjacency matrix
T	the number of temporal segments
S_m	a set of sub-trajectories
D	the embedding dimension of the Transformer input
$\tilde{K}$	the number of predefined reference patterns
z'	projected GPS embedding sequence
R	a road sequence
G	a grid embedding sequence
$\tilde{G}$	a fused grid representation

A.2 Baseline

The baseline methods employed in this paper are detailed as follows:

1. GPS-based trajectory representation learning.

- **Traj2vec** [40]: An RNN-based seq2seq model converts GPS trajectory into a feature sequence to learn trajectory representations.

2. Grid-based trajectory representation learning.

- **TrajCL** [4]: The state-of-the-art grid-based method proposes a set of trajectory augmentations on grid trajectory in free space and dual-feature self-attention to learn grid trajectory representations using contrastive learning with the Transformer.

3. Road-based trajectory representation learning.

- **JCLRNT** [31]: A jointly contrastive learning framework that performs within-road, within-trajectory, and cross-scale road-trajectory contrast, maximizing mutual information to obtain robust representations.
- **START** [18]: A self-supervised trajectory representation model that incorporates temporal regularities and travel semantics using a graph attention network and a time-aware trajectory encoder, trained with span-masked recovery and contrastive learning tasks.
- **JGRM** [29]: A joint GPS–route modeling approach with dual encoders and a cross-modal interaction module, trained in a self-supervised manner via masked language modeling (MLM) and cross-modal matching (CMM) objectives.
- **GREEN** [45]: A multimodal trajectory representation method that jointly leverages grid-based and road-based trajectories,

aligned and fused via contrastive learning and masked reconstruction objectives.

4. Travel Time Estimation Models.

- **DutyTTE** [30]: A reinforcement learning-based framework that enhances the alignment between predicted and actual travel routes, and employs a mixture-of-experts mechanism to quantify uncertainty across different road segments.
- **MULT-TTE** [24]: A multi-faceted model that decomposes routes into trajectory, attribute, and semantic sequences, combines sequential learning with Transformer encoders, and incorporates self-supervised masking and multi-task optimization.

Table 4: The transferability accuracy of MDTI across three cities.

Scenario	MAE↓	RMSE↓	MAPE↓
Porto → Chengdu			
w/ Transfer	1.157	1.634	0.193
w/o Transfer	1.143	1.508	0.183
Chengdu → Xi'an			
w/ Transfer	1.153	1.568	0.173
w/o Transfer	1.169	1.614	0.170

A.3 Trajectory Length Impact (RQ3)

We analyze the effect of fixed and dynamic trajectory lengths on model performance by comparing prediction error metrics. Specifically, we evaluate three fixed lengths (10, 65, and 120) and one dynamic adjustment strategy, using MAE, RMSE, and MAPE as evaluation metrics. The results are presented in Figure 3.

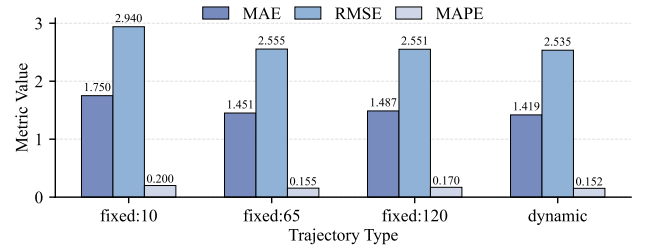


Figure 3: Comparison of fixed and dynamic trajectory lengths on prediction errors.

Under fixed-length conditions, prediction errors decrease as trajectory length increases. This phenomenon indicates that longer trajectories contain a more extensive history of object motion, thereby providing the model with more comprehensive information on motion patterns. In contrast, shorter trajectories provide limited information, making it difficult for the model to distinguish whether an object is moving at a constant speed, accelerating, or turning, which increases the uncertainty of prediction. Furthermore, longer trajectories demonstrate stronger noise resistance. Short trajectories are highly sensitive to measurement noise or transient irregular movements (e.g., obstacle avoidance), where even a single abnormal point can significantly disturb the judgment of the overall trend; meanwhile, longer trajectories, by virtue of incorporating more data points, can effectively smooth out random noise and allow the model to focus on stable and intrinsic motion trends.

More importantly, the dynamic trajectory strategy consistently outperforms all fixed-length configurations across all error metrics. This result demonstrates that adaptive adjustment of the trajectory length allows the model to provide the most appropriate amount of contextual information for different motion scenarios, yielding optimal or near-optimal prediction performance.

A.4 Transferability Study (RQ4)

In Table 4, we evaluate the transferability of our pre-trained model across distinct urban scenarios through a series of experiments. In particular, we fine-tuned the model on the TTE task to adapt it to different city environments, while employing the transferred prompt files during task training.

The results demonstrate that transferring from Porto to Chengdu did not yield additional improvements; instead, the performance slightly declined. Nevertheless, the outcomes remain competitive compared with other baselines. This phenomenon may be attributed to the substantial differences in traffic patterns between the two cities, which limit the effectiveness of transfer learning. Conversely, transferring from Chengdu to Xi'an led to overall performance gains, suggesting that the two cities share more similar traffic characteristics, thereby enabling transfer learning to exert a positive effect.

A.5 Implementation Details

As a supplement to the implementation details in the main paper, we provide a detailed explanation of Algorithm 1, which describes the multimodal pretraining process of MDTL.

In the initialization stage, we set the model parameters Θ (line 1) and prepare the road network graph $\mathcal{G}_{\text{road}}$, the grid graph $\mathcal{G}_{\text{grid}} = (X, A)$, and the GPS trajectories $\{\tau_i\}$. Next, in Stage 1, for each training iteration we sample a mini-batch of trajectories and, for each trajectory τ_i (lines 2-3), construct its road segment sequence on $\mathcal{G}_{\text{road}}$. We then update the road node features with a GAT module and feed the resulting road sequence into a Transformer to obtain the road sequence representation R_i (lines 4-8). Then, in Stage 2, for each trajectory we extract local grid subgraphs along its path on $\mathcal{G}_{\text{grid}}$, apply a two-layer GAT on (X, A) to obtain grid node embeddings H , and project them to form the grid sequence embeddings G_i (lines 9-13). In Stage 3, we further process each GPS trajectory τ_i by segmenting it into sub-trajectories and computing motion patterns. These patterns are organized into structured prompts $\text{Prompt}_{i,t}$, which are fed into a language model $\text{LM}(\cdot)$ to obtain GPS token embeddings $Z_i = \{z_{i,t}\}_{t=1}^T$ (lines 14-18). Next, in Stage

4, we align G_i and Z_i into the same latent space and use a residual fusion module to obtain a fused grid representation $\tilde{G}_i$. We then apply cross-attention between the road sequence representation R_i and $\tilde{G}_i$ to produce the fused trajectory representation $\hat{R}_i$ (lines 19-23). In Stage 5, we compute a contrastive loss $\mathcal{L}_{\text{CL}}$ between the fused grid representations and the road sequence representations (line 24), encouraging trajectories from the same instance to be close while pushing apart those from different instances. Finally, in Stage 6, we randomly mask a subset of road tokens in $\{R_i\}$, predict the masked tokens with the road encoder, and compute the MLM loss $\mathcal{L}_{\text{MLM}}$. The total loss $\mathcal{L} = \mathcal{L}_{\text{CL}} + \mathcal{L}_{\text{MLM}}$ is then used to update the model parameters Θ via gradient descent (lines 25-28), and this process is repeated until convergence.

A.6 Case Study

We randomly selected ride-hailing trips in Porto as our testbed. We simulated travel-time estimation using different approaches and compared the estimated durations with the ground-truth travel times. As shown in Figure 4, we visualize a local region of Porto using three types of base maps: grids, GPS traces, and the road network. We use color-coded arrows to indicate the predictions produced by different models.

Most baselines tend to follow routes dictated by their input trajectory representation when estimating travel time. For example, the GPS-based Traj2vec plans routes according to GPS points, whereas the grid-based TrajCL infers routes based on grid transitions. In contrast, our method jointly leverages multiple trajectory modalities to identify the most plausible route, and its travel-time estimates are the closest to the ground truth. These results demonstrate the effectiveness of the proposed approach.

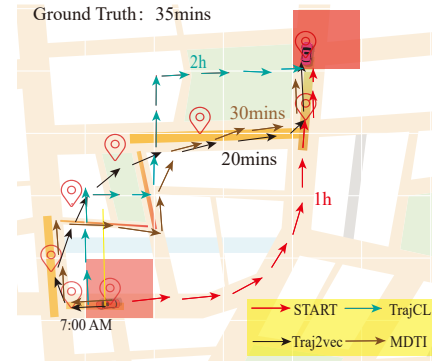


Figure 4: A case study of travel time estimation.

Algorithm 1 Multimodal Pretraining Process

Input: Road network graph $\mathcal{G}_{\text{road}}$, grid graph $\mathcal{G}_{\text{grid}} = (X, A)$, GPS trajectories $\{\tau_i\}$, prototype pattern set $\mathcal{P}$, batch size N_b , learning rate η .

Output: all trained model parameters Θ .

```

1: Initialize parameters  $\Theta$ 
2: for Each training iteration do
3:   Sample a mini-batch of trajectories  $\{\tau_i\}_{i=1}^{N_b}$ 
   Stage 1: Road Trajectory Encoding:
4:   for Each trajectory  $\tau_i$  do
5:     Construct road segment sequence  $r_i$  from  $\tau_i$  on  $\mathcal{G}_{\text{road}}$ 
6:     Update road node features by GAT on  $\mathcal{G}_{\text{road}}$ 
7:     Encode  $r_i$  with Transformer to obtain road sequence representation  $R_i$ 
8:   end for
   Stage 2: Grid Trajectory Encoding:
9:   for each trajectory  $\tau_i$  do
10:    Extract local grid subgraphs along  $\tau_i$  and obtain grid node features  $X$ 
11:    Apply a two-layer GAT on  $(X, A)$  to compute grid node embeddings  $H$ 
12:    Project  $H$  into grid sequence embeddings  $G_i$ 
13:   end for
   Stage 3: GPS Trajectory Encoding:
14:   for Each trajectory  $\tau_i$  do
15:    Segment  $\tau_i$  into sub-trajectories and compute motion patterns
16:    Form structured prompts  $\text{Prompt}_{i,t}$  and feed them into language model  $\text{LM}(\cdot)$ 
17:    Obtain GPS token embeddings  $Z_i = \{z_{i,t}\}_{t=1}^T$ 
18:   end for
   Stage 4: Embedding Fusion:
19:   for Each trajectory  $\tau_i$  do
20:    Align  $G_i$  and  $Z_i$  into the same latent space
21:    Fuse them via residual fusion to get fused grid representation  $\tilde{G}_i$ 
22:    Apply cross-attention between road sequence  $R_i$  and  $\tilde{G}_i$  to obtain fused trajectory representation  $\hat{R}_i$ 
23:   end for
   Stage 5: Contrastive:
24:   Compute contrastive loss  $\mathcal{L}_{\text{CL}}$  between  $\{\tilde{G}_i\}$  and  $\{\hat{R}_i\}$ 
   Stage 6: Masked Language Modeling:
25:   Randomly mask a subset of road tokens in  $\{\hat{R}_i\}$ 
26:   Predict the masked tokens with the road encoder and compute MLM loss  $\mathcal{L}_{\text{MLM}}$ 
27:   Compute total loss  $\mathcal{L} = \mathcal{L}_{\text{CL}} + \mathcal{L}_{\text{MLM}}$ 
28:   Update parameters  $\Theta \leftarrow \Theta - \eta \nabla_{\Theta} \mathcal{L}$ 
29: end for
30: return all trained parameters  $\Theta$ 

```